Real-time Fall Detection Using an Ear-worn Accelerometer: Hybrid Machine Learning Approach and Robust Feature Selection

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***Abstract*—Falls significantly contribute to injury-related hospitalizations among older adults, necessitating reliable real-time detection systems. In this study, we propose an ear-worn, accelerometer-only device for fall detection, prioritizing user comfort and energy efficiency. Utilizing publicly available datasets (SisFall and UMAFall), accelerometer signals were segmented into 4-second windows without any filtering. We initially explored threshold-based methods but excluded them due to excessive false alarms, subsequently focusing on machine learning (ML) approaches. We evaluated two approaches: (i) a continuous ML classifier identifying falls and activities of daily living (ADLs), and (ii) a hybrid method using a threshold-triggered binary classifier for rapid fall identification combined with continuous ADL monitoring. Two sets of handcrafted statistical features (initially 14, expanded to 33, and later optimized) were benchmarked across several ML algorithms. Extra Trees achieved up to 99.9% accuracy on binary classification tasks (fall vs. ADL) using SisFall data, while Linear Discriminant Analysis (LDA) demonstrated strong generalization (96.2%) on unseen ear-level data. Integrating the UMAFall dataset further improved model robustness, highlighting key trade-offs between model complexity, computational load, and cross-placement performance. A detailed list of extracted features is provided in Appendix A.**

I. Introduction

Falls are a major health hazard for the elderly: roughly one-third of people over 65 experience a fall each year, often resulting in fractures or long periods of incapacitation [1]. Prompt fall detection and alerting can greatly reduce the severity of outcomes by preventing "long lie" situations [1], [2]. Wearable devices (e.g. smartphones, pendants, smartwatches) have been widely studied for this purpose, typically using accelerometers (and often gyroscopes) to detect the high accelerations or orientations associated with falls [3], [4]. In this work we propose an ear-worn device (earpiece) with an integrated 3-axis accelerometer (LIS2DW12) to automatically detect falls and notify caregivers. An in-ear sensor has advantages in user acceptance (it resembles a hearing aid) and continuous use, but poses unique challenges since only head accelerations (no gyroscope) are available. Due to the lack of a dedicated ear-worn fall dataset, we use the public SisFall database (waist-mounted IMU) for development. SisFall comprises 19 types of ADLs and 15 types of falls performed by young and older adults [1] and its creators report that even a simple threshold-based classifier can achieve up to 96% fall-detection accuracy on it [1]. We will build on these data with three design concepts (a simple threshold detector, a single ML classifier, and a hybrid threshold+ML approach) and compare their performance.

II. Literature Review

Falls are a significant public health concern, especially among elderly populations, often resulting in severe injuries and long-term consequences. Prompt and reliable fall detection systems are crucial for reducing the severity of outcomes associated with falls, such as prolonged periods of incapacitation due to delayed assistance [1], [2]. Wearable sensor technologies have been increasingly studied for fall detection applications, primarily employing accelerometers, gyroscopes, or combinations thereof, to capture sudden changes in body movements associated with falling events [3], [4].

A. Threshold-based Fall Detection

Threshold-based detectors rely on fixed cutoffs applied to acceleration magnitudes or orientation changes to determine fall events. These approaches are computationally efficient and easy to implement. Aziz and Robinovitch (2011) demonstrated that simple threshold-based systems could achieve high accuracy in controlled environments, with sensitivities around 92–100% and specificities exceeding 95% [3]. However, such systems face significant limitations in real-world applications, as non-fall-related sudden movements (such as quick sitting or jumping) can trigger false alarms. Variations in user behavior and physiology further exacerbate this issue, necessitating adaptive or user-specific threshold calibration [3], [6].

B. Machine Learning-based Fall Detection

Machine learning (ML) methods have emerged as a more robust alternative, capable of learning complex patterns in accelerometer and gyroscope data. Aziz et al. (2016) conducted a comprehensive comparison of threshold-based and ML-based algorithms, reporting superior performance for ML methods, especially support-vector machines (SVM), which provided a better balance of sensitivity and specificity [6]. Yu et al. (2024) utilized Hidden Markov Models and achieved 99.2% sensitivity and 99.0% specificity, demonstrating the superior potential of ML classifiers in accurately distinguishing falls from normal activities [5]. Similarly, Liu et al. (2020) leveraged rich time-domain features from accelerometer signals, achieving up to 97.6% accuracy with SVM classifiers, highlighting the importance of feature selection and extraction in enhancing classifier performance [7].

Further research by Chelli et al. (2020) employed ensemble classifiers and reported accuracies reaching 99.1% using multiple time and frequency-domain features, reinforcing the effectiveness of advanced ML algorithms in fall detection tasks [7]. Although ML models typically outperform simple threshold-based methods, they require increased computational resources and energy consumption, which poses a significant constraint for wearable edge devices.

C. Hybrid Fall Detection Approaches

Hybrid fall detection strategies combine the low computational complexity of threshold-based methods with the accuracy of ML models. Li et al. (2009) developed a hybrid detector integrating accelerometer and gyroscope data, significantly reducing both false positives and negatives compared to accelerometer-only methods [4]. More recently, Wang et al. (2024) introduced a two-tier approach combining threshold-based initial screening with subsequent ML classification, optimizing computational resources by applying intensive processing only to candidate events [5]. This strategy effectively balances detection accuracy and computational efficiency, aligning with the demands of real-time wearable systems.

D. Sensor Placement and Ear-worn Sensors

Sensor placement significantly influences fall detection accuracy. Research consistently indicates that torso-mounted sensors (waist or chest) yield the highest accuracy due to proximity to the body's center of mass [8]. Conversely, peripheral placements such as wrist or ankle sensors often suffer from reduced accuracy. Despite their lower popularity, ear-worn sensors have gained attention for their discreet nature, comfort, and potential for continuous wear, resembling common devices such as hearing aids.

Activity recognition studies using ear-worn accelerometers indicate promising performance despite inherent placement limitations. Skoglund et al. (2021) achieved approximately 84% accuracy for classifying nine activities of daily living (ADLs) using an ear-worn sensor, slightly below the waist-mounted sensor accuracy of around 90% [9]. These findings suggest that, although ear-mounted sensors might face limitations in signal quality and orientation detection compared to torso-mounted sensors, they are still viable for capturing relevant motion dynamics.

Boborzi et al. (2024) conducted further research with ear-worn accelerometers, successfully achieving approximately 95% accuracy across various ADLs, using comprehensive statistical and spectral features [11]. This indicates that carefully selected features and optimized segmentation strategies can compensate for placement-related challenges, supporting the feasibility of ear-level sensors for reliable fall detection applications.

E. Feature Extraction and Selection

Feature extraction plays a critical role in ML-based fall detection systems. Effective feature sets typically include statistical metrics from time and frequency domains, such as mean, variance, standard deviation, root mean square (RMS), range, skewness, kurtosis, signal magnitude area (SMA), zero-crossing rate, FFT coefficients, and spectral entropy [11], [12], [13]. Studies consistently emphasize that combining multiple feature types significantly improves model performance. Bennasar et al. (2022) demonstrated that integrating RMS and spectral features significantly boosted classifier accuracy to around 98%, whereas using only time-domain features resulted in substantially lower accuracy [13].

Feature selection is crucial to enhance model generalizability and computational efficiency. Althobaiti et al. (2020) showed that feature dimensionality reduction, identifying a subset of 13 critical features from an initial set of 72, maintained high accuracy while minimizing redundancy and computational overhead [7]. These insights underline the importance of carefully designing compact yet expressive feature sets, particularly vital for low-power wearable devices such as ear-worn accelerometers.

F. Summary

Previous research clearly indicates that ML and hybrid detection methods outperform traditional threshold-based approaches in accuracy and reliability. Although torso-mounted sensors have typically demonstrated superior performance, recent studies on ear-worn sensors show significant promise, provided that appropriate signal processing and feature extraction methods are utilized. Hybrid systems and optimal feature selection represent a balanced solution, offering computational efficiency and robust detection performance, especially suited for resource-constrained wearable devices. These considerations guided the methodological choices in our study, leveraging comprehensive feature extraction without applying filters to raw acceleration data from ear-worn devices to maximize system robustness and efficiency.

III. Methodology

This section describes our methodological approach in developing and evaluating a robust, real-time fall detection system based on an ear-worn accelerometer. Our primary objective was to minimize false alarms while ensuring high detection accuracy in realistic usage scenarios. Given the limitations associated with threshold-based methods, particularly their tendency to generate numerous false alarms, we excluded them from our practical evaluations. Instead, we focused exclusively on two machine-learning (ML) approaches: a continuous ML classifier and a hybrid ML model.

A. Conceptual Design Models

We adopted two distinct modeling strategies to ensure efficient computation and robust performance:

1) **Continuous ML Classifier (Combined Model)**: This model continuously processes incoming accelerometer data in fixed 4-second windows, classifying each window into one of five predefined activity categories: walking,

jogging, stairs, sitting, and falls. The continuous classification allows immediate and continuous monitoring, facilitating prompt responses upon detecting falls.

2) **Hybrid ML Classifier (Binary + ADL Classifiers)**: The hybrid model utilizes a two-stage approach:
   - *Stage 1 - Threshold Triggered Binary Classification*: We set a predefined acceleration threshold. If the signal surpasses this threshold, the binary classifier analyzes the preceding 4 seconds of data to rapidly determine if the event is a fall or non-fall (ADL).
   - *Stage 2 - Continuous ADL Classification*: Independently and continuously classifies 4-second windows to identify ongoing activities (walking, jogging, stairs, sitting), providing continuous context awareness.

B. Dataset and Preprocessing

Initially, we utilized the widely recognized SisFall dataset, which contains 19 activities (D01–D19) and 15 fall scenarios (F01–F15), recorded using a waist-mounted accelerometer (ADXL345) and gyroscope at a sampling rate of 200 Hz. To enhance generalization and account for sensor placement differences, we later integrated the UMAFall dataset, also collected at 200 Hz. For the SisFall dataset, accelerometer values were converted to g units by dividing by 256, following the ADXL345 datasheet and the SisFall configuration. We also swapped the X and Y axes of the SisFall accelerometer data to better match the orientation of an ear-worn sensor, significantly improving detection accuracy.

C. Signal Segmentation

For SisFall dataset segmentation, we encountered several recordings containing multiple activities within a single session (e.g., "Slowly sit in a half-height chair, wait a moment, and stand up slowly"). To accurately segment these complex recordings, we calculated the combined standard deviation magnitude of the gyroscope's three axes using a sliding window. A dynamic threshold, determined based on the maximum output value, allowed us to identify and extract segments containing significant movements while ignoring idle periods.

This dynamic thresholding approach effectively segmented each activity, enabling precise labeling and improving dataset quality. Additionally, activities involving continuous motion, such as walking, were segmented into shorter, uniform 4-second windows, thus increasing the number of samples. To ensure balanced dataset representation, we specifically selected activities with a higher number of available samples (walking, jogging, stairs, sitting).

When merging the SisFall and UMAFall datasets, we standardized all SisFall signal lengths to match UMAFall's signal lengths, facilitating uniform feature extraction and consistent model training.

D. Feature Extraction and Selection

Initially, we extracted 14 statistical features commonly employed in fall detection literature, primarily inspired by the SisFall dataset. Recognizing the potential for further accuracy improvement, we expanded this initial set to 33 features, enhancing model performance significantly during testing with data from the same dataset.

After merging datasets, we selected a reduced and optimized feature set emphasizing computational efficiency suitable for real-time, wearable implementations. The final selected features included Mean, Variance, Standard Deviation, Median, Maximum/Minimum, Peak-to-Peak Amplitude, and Percentiles, clearly detailed in Table I. Comprehensive descriptions and formulas for all features extracted initially are provided in Appendix A.

TABLE I: Final selected features

Feature	**Formula / Description**
Mean	$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i$
Variance	$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2$
Standard Deviation	$\sigma = \sqrt{\sigma^2}$
Median	$\text{median}(x)$
Maximum / Minimum	$\max(x),\ \min(x)$
Peak-to-Peak Amplitude	$\text{ptp}(x) = \max(x) - \min(x)$
Percentiles	$P_{25}(x),\ P_{75}(x)$

E. Classifier Training and Evaluation

Three distinct classifiers were trained and evaluated:

1) **Continuous ML (Combined Model)**: Trained on combined ADL and fall data, providing immediate, comprehensive activity classification.
2) **Binary Classifier**: Trained specifically for rapid differentiation between fall and ADL events.
3) **ADL Classifier**: Exclusively classifying non-fall ADLs, activated only after the binary classifier identifies a non-fall event.

We systematically evaluated several classical ML algorithms, including Random Forest, Extra Trees, Logistic Regression, Linear Discriminant Analysis (LDA), Support Vector Machines (SVM), and k-Nearest Neighbors (kNN). Model performance was assessed using standard metrics: accuracy, precision, recall, and F1-score, employing stratified 5-fold cross-validation to ensure reliable performance estimation.

F. Limitations and Practical Considerations

Key limitations of our methodology include:

- **Sensor Placement Differences**: Discrepancies between waist- and ear-mounted accelerometer signals may reduce direct generalization accuracy.
- **Lack of Gyroscope Data**: Limited to accelerometer-only data, which restricts orientation detection and potentially limits detection performance.

Despite these constraints, integrating multiple datasets and rigorous feature selection enhanced our model's robustness and practical applicability for real-time, ear-level fall detection.

IV. Results

A. SisFall Dataset Performance

1) 14 feature Classifier Performance: **Combined Classifier (ADLs + Falls)**:

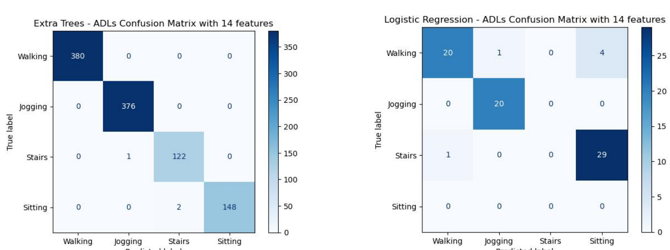
- Best test accuracy: Extra Trees (0.9950), Random Forest (0.9906)
- Unseen data performance was generally poor across most models due to limited information content in the smaller feature set.
- LDA again showed relatively better generalization (0.4937), recognizing walking, jogging, and fall events with higher accuracy.

(a) Test Set (b) Unseen Data

Fig. 1: Combined Classifier Accuracy (14 features) on Test Set and Unseen Data

ADLs-Only Classifier

- Best test accuracy: Extra Trees (0.9971), Random Forest (0.9913)
- Unseen data performance: Logistic Regression had the highest unseen accuracy (0.5333), correctly detecting walking and jogging classes.

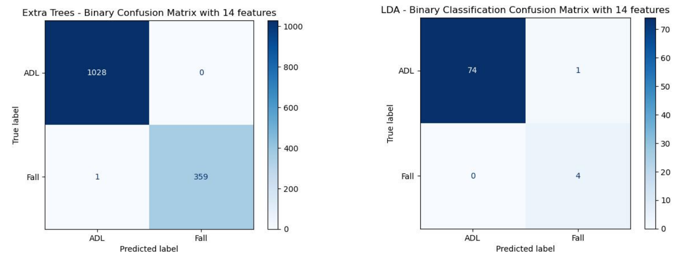


(a) Test Set (b) Unseen Data

Fig. 2: ADLs Classifier Accuracy (14 features) on Test Set and Unseen Data

Binary Classifier (Fall vs. ADL)

- Best test accuracy: Extra Trees (0.9993), Logistic Regression (0.9986), LDA (0.9978)
- Unseen data: LDA excelled again (0.9873), misclassifying only one ADL segment out of 79.

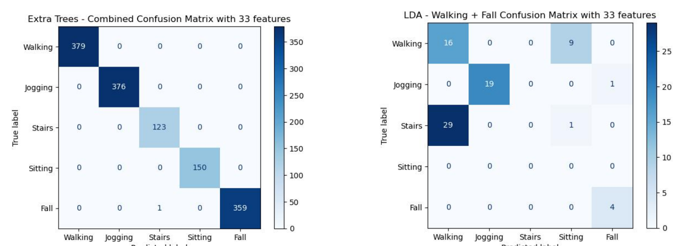


(a) Test Set (b) Unseen Data

Fig. 3: Binary Classifier Accuracy (14 features) on Test Set and Unseen Data

2) 33 feature Classifier Performance: **Combined Classifier (ADLs + Falls)**:

- Best test accuracy: Extra Trees (0.9993), Random Forest (0.9935)
- Unseen data performance was limited due to signal pattern mismatch caused by sensor placement (ear vs. waist). Most models misclassified walking and jogging.
- LDA showed the highest unseen generalization (0.4937), accurately classifying walking, jogging, and fall segments better than others.



(a) Test Set (b) Unseen Data

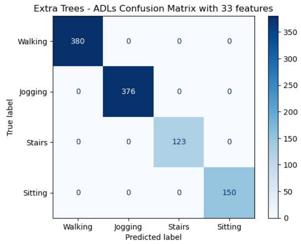
Fig. 4: Combined Classifier Accuracy (33 features) on Test Set and Unseen Data

ADLs-Only Classifier

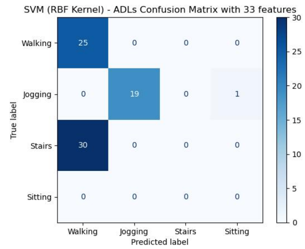
- Best test accuracy: Extra Trees (100%), Random Forest (99.32%)
- Unseen data performance: SVM achieved the best generalization (0.5867), correctly identifying jogging and most walking segments.

Binary Classifier (Fall vs. ADL)

- Best test accuracy: Extra Trees (0.9993), LDA (0.9986), Random Forest (0.9993)
- Unseen data: LDA again demonstrated superior generalization (0.9620), missing only 3 ADL segments out of 79.

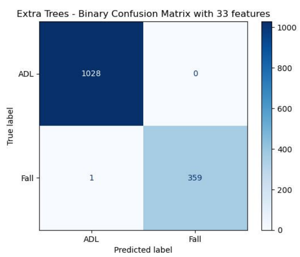


(a) Test Set

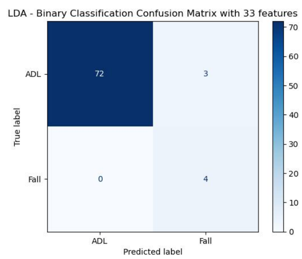


(b) Unseen Data

Fig. 5: ADLs Classifier Accuracy (33 features) on Test Set and Unseen Data



(a) Test Set



(b) Unseen Data

Fig. 6: Binary Classifier Accuracy (33 features) on Test Set and Unseen Data

3) Key Observations: The continuous ML classifier achieved very high accuracy on SisFall data across both 14 and 33 feature sets. Specifically, for binary fall vs. ADL classification, Extra Trees achieved the highest accuracy (99.9%), with Logistic Regression and LDA also performing exceptionally well. However, generalization to unseen ear-level data revealed significant performance differences among models. While Extra Trees and Random Forest excelled on the original dataset, their generalization was limited. LDA consistently provided better performance on unseen data (98.73% for 14 features and 96.20% for 33 features).

The ADLs-only classifiers similarly showed excellent test accuracy (Extra Trees: 99.71% with 14 features, 100% with 33 features). Yet, generalization remained challenging, with Logistic Regression (53.33% with 14 features) and SVM (58.67% with 33 features) performing best.

Table II summarizes the performance of the 14-feature and 33-feature classifiers across different tasks, highlighting the best test and unseen accuracies achieved by each model. The results indicate that while the 33-feature model generally performs better on the test set, the 14-feature model shows improved unseen data generalization for binary classification tasks.

TABLE II: Feature Size Comparison: 14 vs. 33 Features

Task Type	Best Test Accuracy (33F)	Best Test Accuracy (14F)	Best Unseen Accuracy (33F)	Best Unseen Accuracy (14F)
Combined (5-class)	(Extra Trees) 0.9993	(Extra Trees) 0.9950	(LDA) 0.4937	(LDA) 0.4937
ADLs (4-class)	(Extra Trees) 1.0000	(Extra Trees) 0.9971	(SVM) 0.5867	(Logistic Reg.) 0.5333
Binary (2-class)	(Extra Trees) 0.9993	(Extra Trees) 0.9993	(LDA) 0.9620	(LDA) 0.9873

B. Integration of SisFall and UMAFall Datasets

Merging the UMAFall dataset with SisFall substantially improved model generalization on ear-level accelerometer signals. After integrating both datasets and standardizing the signal lengths, we employed statistical feature selection, yielding a compact and robust feature set. Models retrained with the enriched dataset exhibited significant improvements:

- **Combined Classifier (ADLs + Falls)**: Highest generalization performance was observed for LDA, MLP, and Random Forest.
- **ADL-Only Classifier**: LDA, MLP, and Logistic Regression showed superior generalization capabilities.
- **Binary Classifier (Fall vs. ADL)**: Logistic Regression, SVM, and Random Forest provided the best unseen data accuracy.

Overall, these improvements confirmed the critical role of dataset diversity and effective feature selection in enhancing model robustness for real-world applications.

C. SisFall and UMAFall Dataset Integration

To enhance the robustness of our fall detection system, we integrated the **UMAFall** dataset into our training process. This dataset contains similar activities of daily living (ADLs) and falls, recorded at 200 Hz, but with different sensor placements and user demographics. The integration aimed to improve generalization capabilities on real-world signals recorded from the **earpiece**.

To improve generalization on real-world signals recorded from the **earpiece**, we addressed the observed issue of **low accuracy on unseen data**—primarily caused by the mismatch between the waist-worn sensors in the SisFall dataset and the ear placement of our actual device.

1) Feature Selection and UMAFall Integration: We enhanced the training dataset by integrating samples from the **UMAFall** dataset, which includes similar ADLs and falls captured at 200 Hz. While UMAFall is also not ear-specific, its inclusion introduced additional variation in signal patterns and motion dynamics, aiding the generalization capability of our models.

2) Feature Selection Strategy: From the original 33 extracted features, we applied **statistical feature selection** to remove redundant or non-informative components. The selected features included descriptive statistics computed per

TABLE III: Final selected features

Feature	Formula / Description
Mean	$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i$
Variance	$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2$
Standard Deviation	$\sigma = \sqrt{\sigma^2}$
Median	$\text{median}(x)$
Maximum / Minimum	$\max(x),\ \min(x)$
Peak-to-Peak Amplitude	$\text{ptp}(x) = \max(x) - \min(x)$
Percentiles	$P_{25}(x),\ P_{75}(x)$

signal axis (X, Y, Z), and optionally on their vector magnitude $\mathbf{mag} = \sqrt{x^2 + y^2 + z^2}$:

These features were extracted from each of the 3 axes (X, Y, Z) and optionally from the magnitude channel. This resulted in a compact yet expressive feature vector that is computationally efficient and robust to orientation variance.

3) Performance Improvements: After applying feature selection and enriching the dataset with UMAFall signals, we observed significant accuracy improvements on **unseen earpiece test data**.

Model-wise performance gains:

- **Combined Classifier (ADLs + Falls)**: Best performance from **LDA**, **MLP**, and **Random Forest**.
- **ADL-Only Classifier**: Top accuracy achieved by **LDA**, **MLP**, and **Logistic Regression**.
- **Binary Classifier (Fall vs. ADL)**: Best models were **Logistic Regression**, **SVM**, and **Random Forest**.

These results confirm that **feature quality and dataset diversity** play a more critical role than increasing model complexity when designing robust fall detection systems for wearable edge devices.

V. Discussion

Our study underscores essential considerations for practical fall detection systems using wearable accelerometers. Although tree-based classifiers consistently exhibited superior accuracy on dataset-specific tests, their generalization to novel sensor placements proved limited. This limitation highlights the importance of selecting models that generalize well across diverse sensor orientations. Linear models, particularly LDA, demonstrated greater robustness to sensor placement variations, making them favorable for practical deployment.

The hybrid ML approach, combining rapid binary classification triggered by threshold events with continuous ADL monitoring, effectively balanced computational efficiency and detection accuracy. Excluding pure threshold-based methods due to high false alarm rates proved justified, given the improved accuracy and reduced false positives of the hybrid approach.

Integration of UMAFall with SisFall introduced valuable variability, significantly enhancing generalization capabilities. This finding emphasizes the necessity of diverse training

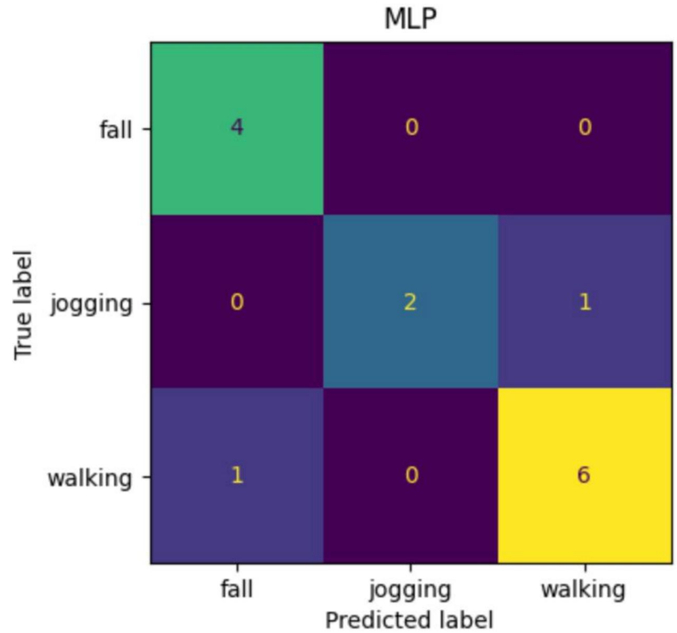


Fig. 7: Enter Caption

datasets representative of real-world usage conditions. Moreover, our optimized feature set maintained computational efficiency without sacrificing discriminative power, further validating the practicality of the selected features.

Limitations, such as the absence of gyroscopic data and inherent sensor-placement discrepancies, remain critical considerations. Future research should explore multi-modal sensor integration and advanced machine learning methods, including deep learning, to potentially overcome these limitations and further enhance fall detection performance.

VI. Conclusion

In this research, we developed and evaluated a practical, real-time fall detection system employing an ear-worn accelerometer. By systematically exploring ML-based continuous and hybrid classification methods, we demonstrated the importance of careful feature selection, dataset integration, and sensor orientation considerations. Our results indicate that while tree-based models achieve excellent accuracy on familiar data, linear models like LDA provide superior generalization in practical, real-world scenarios. Integrating diverse datasets such as UMAFall significantly improved model robustness, underscoring dataset diversity's importance. Our compact, optimized feature set ensures computational efficiency suitable for real-time wearable implementation. Future work should focus on incorporating additional sensor modalities and advanced ML approaches to further improve reliability and accuracy in practical fall detection applications.

VII. Future Work and Suggestions

Future research should explore several promising avenues to further enhance the reliability and accuracy of ear-worn fall detection systems:

- **Sensor Fusion**: Integration of additional sensor modalities, such as gyroscopes, magnetometers, and barometric pressure sensors, to provide complementary data for improved orientation detection and enhanced robustness against false alarms.
- **Advanced Machine Learning Techniques**: Exploration of deep learning approaches, particularly convolutional and recurrent neural networks, to potentially capture complex temporal and spatial patterns within accelerometer signals more effectively.
- **Real-world Validation**: Conducting extensive field testing with diverse user demographics and various real-life scenarios to validate and fine-tune the model's performance and ensure practicality and reliability in actual deployment environments.
- **Personalized Models**: Investigating personalized or adaptive model training methods that can dynamically adjust to individual user behavior, physiology, and daily routines, potentially enhancing accuracy and user acceptance.
- **Energy-efficient Implementation**: Developing and optimizing algorithms specifically designed for low-power embedded systems to ensure prolonged battery life and user comfort in continuous use scenarios.

Addressing these areas will significantly contribute to advancing wearable fall detection technology, ensuring better protection and improved quality of life for older adults and other vulnerable populations.

APPENDIX

TABLE IV: Summary of 33 handcrafted features extracted from each 4-second window

Feature(s)	**Name / Type**	**Description / Formula**
C1–C2	RMS Total / RMS Horizontal	RMS $= \sqrt{\frac{1}{N}\sum_{i=1}^{N} a_i^2}$ (total or horizontal axes)
C3	Peak-to-Peak RMS	$\sqrt{(\max(a) - \min(a))}$
C4–C5	Tilt / Trunk Orientation	Mean tilt angle: $\arctan\left(\frac{\sqrt{x^2+z^2}}{-y}\right)$
C6	Orientation Change	Temporal smoothness: $\text{mean}(x_{t-1}) \cdot \text{mean}(x_t)$
C7	Jerk	Jerk $= \frac{da}{dt} \approx \text{mean}(\lvert\Delta a/\Delta t\rvert)$
C8–C9	Std Dev (Horizontal / 3D)	$\sqrt{\text{std}^2(x) + \text{std}^2(z)}$ and 3D variant
C10	SMA (Signal Magnitude Area)	$\frac{1}{N}\sum(\lvert x\rvert + \lvert y\rvert + \lvert z\rvert)$
C11	Horizontal SMA	$\frac{1}{N}\sum(\lvert x\rvert + \lvert z\rvert)$
C12	Total Magnitude	$\sum\sqrt{x^2+y^2+z^2}$
C13	Horizontal Magnitude	$\sum\sqrt{x^2+z^2}$
C14	Velocity Approximation	Velocity $= \frac{1}{N} \cdot \sqrt{(\sum \text{cumsum}(x))^2 + (\sum \text{cumsum}(z))^2}$
C15–C17	Mean (X, Y, Z)	Per-axis mean acceleration
C18–C20	Std (X, Y, Z)	Per-axis standard deviation
C21–C23	Skewness (X, Y, Z)	Asymmetry in axis distributions
C24–C26	Kurtosis (X, Y, Z)	Peakedness / tail heaviness of signals
C27	Zero-Crossings	Number of crossings in $(a - \bar{a})$
C28–C30	*[Optional]* Frequency Features	Entropy, dominant frequency, and band power (0–5 Hz)
C31–C33	Axis Correlations	Pearson correlation: $\rho(x,y), \rho(y,z), \rho(x,z)$

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